



NORSK DATA

PRODUCT INFORMATION

Product name

COSMOS Telnet/FTP Clients

Product number

211154D

DESCRIPTION OF PRODUCT

This product allows Terminal emulation and file access between an ND machine and remote systems running TCP/IP protocols over Ethernet.

REASON(S) FOR NEW PRODUCT OR VERSION

Version D01

- This product can run over COSMOS TCP/IP Gateway in the Ethernet II controller and over TCP/IP Basic Module for ND-5000 in the Ethernet III controller.
- The Telnet Client in COSMOS Telnet/FTP Clients D01 is compatible with Telnet in Berkely Unix 4.3. It is not compatible with Telnet in Berkely Unix 4.2. Known dependencies are listed on page 3, other dependencies.
- A Domain Name Service (DNS) Resolver is included in these clients. This allows you to access a Domain Name Server when looking up IP-adresses (instead of using the local (SYSTEM)AIP-HOSTS:SYMB file), provided that the (SYSTEM)AIP-RESOLVER:SYMB file exists and contains at least one 'nameserver' entry.

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1 PREREQUISITES

The following things are required in order to use this product.

1.1 HARDWARE PREREQUISITES

The following hardware must be present for this product to be used:

CPU type (any of the following):

- 100 CX
- 110 CX
- 120 CX
- 500
- 5000

Floating format: *Any*

Other hardware:

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None

1.2 SOFTWARE PREREQUISITES

The following software must be installed on the system for this product to be used:

Operating system:

Name

Version/Revision

SINTRAN III L06 Patch file >= 2000B

Other software:

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TCP/IP Basic Module/III

211327 >= B05

OR

COSMOS TCP/IP Gateway for Ethernet

211185 >= C07

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211154D**1.3 OTHER PREREQUISITES**

The following are necessary for this product to be used.

Minimum Permanent Mass Storage:

User	Space (pages)	Number of files
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Any User	250	3
----------	-----	---

	Minimum	Maximum
Number of segments (ND-100)	0	0

Number of RT descriptions	0	0
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Space required on segment files: 0 pages

Requirements regarding internal devices: None.

Requirements regarding batch processors: None.

2 DEPENDENCIES

In addition to the above listed hardware and/or software prerequisites, the following dependencies exist between this and other products. The hardware and software listed here may, for example, be necessary to use the product's full functionality.

2.1 HARDWARE DEPENDENCIES

There are dependency relationships between this product and the following hardware:

None.

2.2 SOFTWARE DEPENDENCIES

There are dependency relationships between this product and the following software:

None.

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2.3 OTHER DEPENDENCIES

In addition to the hardware and software dependency relationships listed above, you should be aware of the following dependencies:

To utilize the DNS Resolver at least one Domain Name Server must be available on the net.

The Telnet Client version D01 included in this product is according to the protocol in version 4.3 of Berkely Unix.

The Telnet Client version D01 is not compatible with:

Telnet version 4.2 of Berkeley Unix.

3 DOCUMENTATION

Included (distributed free with product)

TitleNumber

COSMOS Telnet/FTP Client User Guide

ND-860284EN 1

Optional (must be purchased)

TitleNumber

OpenLAN Network Supervisor Guide

ND-830107EN 3

OpenLAN DNS Operation Guide

ND-830145EN 1

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4 INSTALLATION PROCEDURE

4.1 AUTOMATIC INSTALLATION

None.

4.2 NON-AUTOMATIC INSTALLATION

Insert the floppy and enter directory:

@ENTER-DIRECTORY 211154D-XX-01D,floppy-drive-<no.>,<unit>

Copy the files FTP-CLIENT-D:PROG, TELNET-CLIENT-D:PROG and RSH-CLIENT-D:PROG to a user of your choice with public read access, and dump them reentrant:

@DUMP-PROGRAM-REENT TELNET-CLIENT-D (<USERNAME>)TELNET-CLIENT-D:PROG
@DUMP-PROGRAM-REENT FTP-CLIENT-D (<USERNAME>)FTP-CLIENT-D:PROG
@DUMP-PROGRAM-REENT RSH-CLIENT-D (<USERNAME>)RSH-CLIENT-D:PROG

If name-to-address resolution is to be performed by a domain name server, copy the AIP-RESOLVER:SYMB file to user SYSTEM with public read access, and edit it to include the local default 'DOMAIN' name (to be added to all non-fully-qualified domain names), and the IP-address of at least one 'NAMESERVER' on the net.

If TCP/IP is on a remote computer, copy the files: AIP-HOSTS:SYMB, AIP-SERVICES:SYMB, AIP-NETWORKS:SYMB, and AIP-PROTOCOL:SYMB to user SYSTEM with public read access and then define *TCPGATE in XMSG where the servers are installed. (Note that TCP/IP can only be on a remote computer if you are using the COSMOS TCP/IP Gateway product).

Add your own internet-address to the AIP-HOSTS:SYMB file using an editor.

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5 MODIFICATIONS

5.1 NEW FUNCTIONS/COMMANDS

A RSH-client has been included in this product. It may be used to perform simple one-line command(-pipe)s on (Unix-)systems that fully support the Remote Shell protocol. This protocol does not rely on using passwords, but requires that every system and user that is to be allowed to execute commands from remote must be known in advance.

Thus, if user (account) 'me-too' on system 'our-sintran-sys' is to be allowed to execute commands on system 'our-uniline-sys' as user 'anyone' whose home directory is '/usr/anyone' and default shell is the Bourne shell, the file '/usr/anyone/.rhosts' must exist and contain the line 'our-sintran-sys me-too', and the '/etc/hosts' file must contain a line for the IP-address of 'our-sintran-sys' and give this name as the first name for that address.

Provided that all the conditions specified above are met, the RSH-client program could be used to e.g. check the date according to 'our-uniline-sys'. Thus (assuming logged in as 'me-too'):

```
OUR-SINTRAN-SYS@rsh-client our-uniline-sys anyone date<cr>
```

would print a string similar to 'Wed Apr 18 14:16:13 EET 1990', and then leave you with the 'OUR-SINTRAN-SYS@' prompt. A somewhat more complex example of using the RSH-client could be:

```
@rsh our-uniline-sys anyone > "rsh:man" man rshd ^pr -te<cr>
```

which would use the remote on-line MANual facility to see whatever info is available on the RSH Daemon, pipe this thru the PRint filter so as to Expand input tabs to spaces, return the expanded text to the local machine and finally store the text on the new local rsh:man file instead of printing it on the screen.

Note that RSH-client cannot be used to execute interactive commands on a remote system, since all input have to be provided on the command line.

Use @rsh-client<cr><cr> to get help with the program syntax, and refer to any on- or off-line Unix documentation available to you for in-depth information on the Remote Shell protocol in general and topics specific for a particular RSH-server implementation.

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5.2 CHANGED FUNCTIONS/COMMANDS

None.

5.3 REMOVED FUNCTIONS/COMMANDS

None.

5.4 OTHER MODIFICATIONS

A DNS Resolver is included in all clients. If the (SYSTEM)AIP-RESOLVER:SYMB file does not exist, or contains no 'nameserver' entries, the Resolver code will not be used, and the client will then access the (SYSTEM)AIP-HOSTS:SYMB file just like previous versions. However, if a 'nameserver' entry does exist, a name-to-address resolution request will be sent to the DNS Server at the IP-address specified in the 'nameserver' entry.

6 ERRORS CORRECTED

- Telnet client will now use local echo by default, and handle the echo option processing correctly.
- Telnet client will strip 0 after <CR> when not doing binary.
- Telnet client writes the error message: "Bad character after IAC" only if debug is turned on.
- Ftp client can handle the error message when the connection is idle more than 90 seconds.
- Ftp client now allows any number of arguments to the QUOTE command.
- Ftp client will strip the file version when doing MPUT with file-name conversion ON, unless version is different from one, in which case the semicolon is converted to a hyphen.
- Ftp client will convert characters that are illegal in S-3 file-names to hyphens when doing MGET with file-name conversion ON.

7 ERRORS KNOWN BUT NOT CORRECTED

None.